Example 8.18 A bag contains 5 blue balls and 4 green balls. A ball is drawn at random from the bag. Find the probability that the ball drawn is (i) blue (ii) not blue.

Solution Total number of possible outcomes $n(\mathrm{S}) = 5 + 4 = 9$

(i) Let A be the event of getting a blue ball.
Number of favourable outcomes for the event A. Therefore, $n(\mathrm{A}) = 5$

Probability that the ball drawn is blue. Therefore, $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{5}{9}$

(ii) $\bar{A}$ will be the event of not getting a blue ball. So $P(\bar{A}) = 1 - P(A) = 1 - \dfrac{5}{9} = \dfrac{4}{9}$

Example 8.19 Two dice are rolled. Find the probability that the sum of outcomes is (i) equal to 4 (ii) greater than 10 (iii) less than 13

Solution When we roll two dice, the sample space is given by

$$S = \{\ (1,1),(1,2),(1,3),(1,4),(1,5),(1,6)$$
$$(2,1),(2,2),(2,3),(2,4),(2,5),(2,6)$$
$$(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)$$
$$(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)$$
$$(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)$$
$$(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\ \};\ n(S) = 36$$

(i) Let A be the event of getting the sum of outcome values equal to 4.
Then $A = \{(1,3),(2,2),(3,1)\};\ n(A) = 3$.

Probability of getting the sum of outcomes equal to 4 is $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{3}{36} = \dfrac{1}{12}$

(ii) Let B be the event of getting the sum of outcome values greater than 10.
Then $\mathrm{B} = \{(5,6),(6,5),(6,6)\};\ n(B) = 3$

Probability of getting the sum of outcomes greater than 10 is $P(B) = \dfrac{n(B)}{n(S)} = \dfrac{3}{36} = \dfrac{1}{12}$

(iii) Let C be the event of getting the sum of outcomes less than 13. Here all the outcomes have the sum value less than 13. Hence $C = S$.
Therefore, $n(C) = n(S) = 36$

Probability of getting the total value less than 13 is $P(C) = \dfrac{n(C)}{n(S)} = \dfrac{36}{36} = 1$.

Example 8.20 Two coins are tossed together. What is the probability of getting different faces on the coins?

Solution When two coins are tossed together, the sample space is

$S = \{HH, HT, TH, TT\};\ n(S) = 4$

Let A be the event of getting different faces on the coins.

$A = \{HT, TH\};\qquad n(A) = 2$

Probability of getting different faces on the coins is $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{2}{4} = \dfrac{1}{2}$

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